

# Mathematical Statistics Assignment 5

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### 5.2.3

Let  $X_1, \dots, X_n$  be a random sample from a uniform distribution on the interval  $(\theta - 1, \theta + 1)$ .

- (a) Find a moment estimator for  $\theta$ .
- (b) Use the following data to obtain a moment estimate for  $\theta$ :

11.72, 12.81, 12.09, 13.47, 12.37.

### Solution:

Let  $\sum X_i \sim U(\theta - 1, \theta + 1)$ .

By definition:

$$f(x) = \begin{cases} \frac{1}{\theta+1-(\theta-1)}, & \text{if } \theta - 1 < x < \theta + 1, \\ 0, & \text{otherwise.} \end{cases}$$

### (a) Moment Estimator for $\theta$

By definition:

$$E(\bar{X}) = \frac{\theta - 1 + \theta + 1}{2}.$$

$$E(\bar{X}) = \theta.$$

Thus:

$$\hat{\theta} = \bar{X}.$$

### (b) Moment Estimate for $\theta$

Using the data:

$$\hat{\theta} = \bar{X} = \frac{11.72 + 12.81 + 12.09 + 13.47 + 12.37}{5}.$$

$$\hat{\theta} = 12.48.$$

### 5.2.7

Let  $X_1, \dots, X_n$  be a random sample from a population with pdf:

$$f(x) = \begin{cases} \frac{2\alpha^2}{x^3}, & x \geq \alpha, \\ 0, & \text{otherwise.} \end{cases}$$

Find a method of moments estimator for  $\alpha$ .

## Solution:

To find a method of moments estimator for  $\alpha$ , we proceed as follows:

### Step 1: Probability Density Function (PDF)

The given PDF is:

$$f(x) = \begin{cases} \frac{2\alpha^2}{x^3}, & x \geq \alpha, \\ 0, & \text{otherwise.} \end{cases}$$

This defines a continuous random variable  $X$  for  $x \geq \alpha$ , where  $\alpha > 0$  is the parameter to be estimated.

### Step 2: Calculate the Expected Value

Using the method of moments, we equate the theoretical mean  $\mathbb{E}[X]$  of the distribution to the sample mean  $\bar{X}$ . The mean  $\mathbb{E}[X]$  is:

$$\mathbb{E}[X] = \int_{\alpha}^{\infty} x f(x) dx.$$

Substitute  $f(x) = \frac{2\alpha^2}{x^3}$ :

$$\mathbb{E}[X] = \int_{\alpha}^{\infty} x \cdot \frac{2\alpha^2}{x^3} dx = 2\alpha^2 \int_{\alpha}^{\infty} \frac{1}{x^2} dx.$$

The integral is:

$$\int_{\alpha}^{\infty} \frac{1}{x^2} dx = \left[ -\frac{1}{x} \right]_{\alpha}^{\infty} = \left( 0 - \left( -\frac{1}{\alpha} \right) \right) = \frac{1}{\alpha}.$$

Thus:

$$\mathbb{E}[X] = 2\alpha^2 \cdot \frac{1}{\alpha} = 2\alpha.$$

### Step 3: Equate to the Sample Mean

The sample mean  $\bar{X}$  is:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Using the method of moments:

$$\mathbb{E}[X] = \bar{X}.$$

Substitute  $\mathbb{E}[X] = 2\alpha$ :

$$2\alpha = \bar{X}.$$

Solve for  $\alpha$ :

$$\hat{\alpha} = \frac{\bar{X}}{2}.$$

## Final Answer

The method of moments estimator for  $\alpha$  is:

$$\hat{\alpha} = \frac{\bar{X}}{2}.$$

### 5.2.10

Let  $X_1, \dots, X_n$  be a random sample from a distribution with the pdf:

$$f(x) = \begin{cases} \frac{2\beta-2x}{\beta^2}, & 0 < x < \beta, \\ 0, & \text{otherwise.} \end{cases}$$

Use the method of moments to obtain an estimator of  $\beta$ .

### Solution:

To find the method of moments estimator for  $\beta$ , we proceed as follows:

#### Step 1: Probability Density Function (PDF)

The given PDF is:

$$f(x) = \begin{cases} \frac{2\beta-2x}{\beta^2}, & 0 < x < \beta, \\ 0, & \text{otherwise.} \end{cases}$$

This describes a random variable  $X$  defined on the interval  $(0, \beta)$ .

#### Step 2: Find the Expected Value $\mathbb{E}[X]$

The expected value is given by:

$$\mathbb{E}[X] = \int_0^\beta x f(x) dx.$$

Substitute  $f(x) = \frac{2\beta-2x}{\beta^2}$ :

$$\mathbb{E}[X] = \int_0^\beta x \cdot \frac{2\beta-2x}{\beta^2} dx.$$

Expand and simplify:

$$\mathbb{E}[X] = \frac{1}{\beta^2} \int_0^\beta (2\beta x - 2x^2) dx.$$

Split the integral:

$$\mathbb{E}[X] = \frac{1}{\beta^2} \left[ 2\beta \int_0^\beta x \, dx - 2 \int_0^\beta x^2 \, dx \right].$$

Compute each term: 1. For  $\int_0^\beta x \, dx$ :

$$\int_0^\beta x \, dx = \left[ \frac{x^2}{2} \right]_0^\beta = \frac{\beta^2}{2}.$$

2. For  $\int_0^\beta x^2 \, dx$ :

$$\int_0^\beta x^2 \, dx = \left[ \frac{x^3}{3} \right]_0^\beta = \frac{\beta^3}{3}.$$

Substitute back:

$$\mathbb{E}[X] = \frac{1}{\beta^2} \left[ 2\beta \cdot \frac{\beta^2}{2} - 2 \cdot \frac{\beta^3}{3} \right].$$

Simplify:

$$\mathbb{E}[X] = \frac{1}{\beta^2} \left[ \beta^3 - \frac{2\beta^3}{3} \right].$$

$$\mathbb{E}[X] = \frac{1}{\beta^2} \cdot \frac{\beta^3}{3} = \frac{\beta}{3}.$$

### Step 3: Method of Moments

Using the method of moments, equate  $\mathbb{E}[X]$  to the sample mean  $\bar{X}$ :

$$\frac{\beta}{3} = \bar{X}.$$

Solve for  $\beta$ :

$$\hat{\beta} = 3\bar{X}.$$

### Final Answer

The method of moments estimator for  $\beta$  is:

$$\hat{\beta} = 3\bar{X},$$

where  $\bar{X}$  is the sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

### 5.2.14

Let  $X_1, \dots, X_n$  be a random sample recorded as heads or tails resulting from tossing a coin  $n$  times with unknown probability  $p$  of heads.

1. Find the MLE  $\hat{p}$  of  $p$ .
2. Using the invariance property, obtain an MLE for  $q = 1 - p$ .

How would you use the results you have obtained?

### Solution:

To find the Maximum Likelihood Estimator (MLE) for  $p$  and use the invariance property to find the MLE for  $q = 1 - p$ , we proceed as follows:

#### Step 1: Define the Likelihood Function

Each coin toss is modeled as a Bernoulli trial with probability  $p$  of heads. Let  $X_1, X_2, \dots, X_n$  represent the outcomes of  $n$  independent tosses, where  $X_i = 1$  for heads and  $X_i = 0$  for tails. The probability mass function for each  $X_i$  is:

$$P(X_i = x_i) = p^{x_i}(1-p)^{1-x_i}, \quad x_i \in \{0, 1\}.$$

The likelihood function for the entire sample is:

$$L(p) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i}.$$

Simplify the product:

$$L(p) = p^{\sum x_i}(1-p)^{n-\sum x_i}.$$

Let  $S = \sum_{i=1}^n X_i$ , the total number of heads. The likelihood function becomes:

$$L(p) = p^S(1-p)^{n-S}.$$

#### Step 2: Log-Likelihood Function

Take the natural logarithm of the likelihood function:

$$\ell(p) = \ln L(p) = S \ln p + (n - S) \ln(1 - p).$$

### Step 3: Differentiate the Log-Likelihood

Differentiate  $\ell(p)$  with respect to  $p$ :

$$\frac{d\ell(p)}{dp} = \frac{S}{p} - \frac{n-S}{1-p}.$$

Set the derivative to zero to find the critical point:

$$\frac{S}{p} = \frac{n-S}{1-p}.$$

Simplify:

$$S(1-p) = (n-S)p.$$

Solve for  $p$ :

$$p = \frac{S}{n}.$$

### Step 4: Maximum Likelihood Estimator (MLE)

The MLE for  $p$  is:

$$\hat{p} = \frac{\sum_{i=1}^n X_i}{n}.$$

### Step 5: MLE for $q = 1 - p$

Using the invariance property of MLEs, the MLE for  $q = 1 - p$  is:

$$\hat{q} = 1 - \hat{p}.$$

Substitute  $\hat{p}$ :

$$\hat{q} = 1 - \frac{\sum_{i=1}^n X_i}{n}.$$

### Step 6: How to Use the Results

- **MLE for  $p$ :** Use  $\hat{p}$  to estimate the probability of heads based on the observed data.
- **MLE for  $q$ :** Use  $\hat{q}$  to estimate the probability of tails based on the observed data.

The results can be applied to predict the likelihood of future outcomes for a similar coin or validate the fairness of the coin.

### Final Answer

1. The MLE for  $p$  is:

$$\hat{p} = \frac{\sum_{i=1}^n X_i}{n}.$$

2. The MLE for  $q = 1 - p$  is:

$$\hat{q} = 1 - \frac{\sum_{i=1}^n X_i}{n}.$$

### 5.2.15

Let  $X$  be a random variable representing the time between successive arrivals at a checkout counter in a supermarket. The values of  $X$  in minutes (rounded to the nearest minute) are:

1, 2, 3, 7, 11, 4, 13, 12, 7, 3, 2, 11, 7, 2.

Assume that the PDF of  $X$  is:

$$f(x) = \frac{1}{\theta} e^{-x/\theta}, \quad x \geq 0.$$

Use these data to find the Maximum Likelihood Estimator (MLE)  $\hat{\theta}$ .

How can you use this estimate you have just derived?

Also, find the method of moments estimate.

### Solution:

Let  $X$  be a random variable representing the time between successive arrivals at a checkout counter. The PDF of  $X$  is given by:

$$f(x) = \frac{1}{\theta} e^{-x/\theta}, \quad x \geq 0.$$

The observed data is:

1, 2, 3, 7, 11, 4, 13, 12, 7, 3, 2, 11, 7, 2.

### Step 1: Maximum Likelihood Estimator (MLE)

#### Likelihood Function

For  $n$  independent and identically distributed (i.i.d.) observations  $X_1, X_2, \dots, X_n$ , the likelihood function is:

$$L(\theta) = \prod_{i=1}^n f(X_i) = \prod_{i=1}^n \frac{1}{\theta} e^{-X_i/\theta}.$$

Simplify the product:

$$L(\theta) = \frac{1}{\theta^n} \exp\left(-\frac{\sum_{i=1}^n X_i}{\theta}\right).$$



### Log-Likelihood Function

The log-likelihood function is:

$$\ell(\theta) = \ln L(\theta) = -n \ln \theta - \frac{\sum_{i=1}^n X_i}{\theta}.$$

### Differentiation and Maximization

Differentiate  $\ell(\theta)$  with respect to  $\theta$ :

$$\frac{d\ell(\theta)}{d\theta} = -\frac{n}{\theta} + \frac{\sum_{i=1}^n X_i}{\theta^2}.$$

Set  $\frac{d\ell(\theta)}{d\theta} = 0$  to find the critical point:

$$\frac{n}{\theta} = \frac{\sum_{i=1}^n X_i}{\theta^2}.$$

Simplify:

$$\theta = \frac{\sum_{i=1}^n X_i}{n}.$$

Thus, the MLE for  $\theta$  is:

$$\hat{\theta}_{\text{MLE}} = \bar{X} = \frac{\sum_{i=1}^n X_i}{n}.$$

## Step 2: Method of Moments (MoM)

### Theoretical Mean

For an exponential distribution, the theoretical mean is:

$$\mathbb{E}[X] = \theta.$$

### Equating to the Sample Mean

The method of moments equates the theoretical mean to the sample mean:

$$\mathbb{E}[X] = \bar{X}.$$

Thus, the MoM estimator for  $\theta$  is:

$$\hat{\theta}_{\text{MoM}} = \bar{X}.$$

### Step 3: Compute $\hat{\theta}$ Using the Data

The given data is:

$$1, 2, 3, 7, 11, 4, 13, 12, 7, 3, 2, 11, 7, 2.$$

Compute the sample mean:

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}.$$

The total sum is:

$$\sum_{i=1}^{14} X_i = 84.$$

Thus:

$$\bar{X} = \frac{84}{14} = 6.$$

### Final Answer

1. The Maximum Likelihood Estimator (MLE) for  $\theta$  is:

$$\hat{\theta}_{\text{MLE}} = 6.$$

2. The Method of Moments (MoM) estimator for  $\theta$  is:

$$\hat{\theta}_{\text{MoM}} = 6.$$

### 5.3.7

Let  $X_1, \dots, X_n$  be a random sample from a  $U(0, \theta)$  distribution. Let  $Y_n = \max\{X_1, \dots, X_n\}$ . We know (from Example 5.3.4) that  $\hat{\theta}_1 = Y_n$  is an MLE of  $\theta$ .

- (a) Show that  $\hat{\theta}_2 = 2\bar{X}$  is a method of moments estimator.
- (b) Show that  $\hat{\theta}_1$  is a biased estimator and  $\hat{\theta}_2$  is an unbiased estimator of  $\theta$ .
- (c) Show that  $\hat{\theta}_3 = \frac{n+1}{n}\hat{\theta}_1$  is an unbiased estimator of  $\theta$ .

### Solution:

Let  $X_1, \dots, X_n$  be a random sample from a  $U(0, \theta)$  distribution. Let  $Y_n = \max\{X_1, \dots, X_n\}$ .

**Part (a): Show that  $\hat{\theta}_2 = 2\bar{X}$  is a method of moments estimator**

1. For a  $U(0, \theta)$  distribution, the theoretical mean is:

$$\mathbb{E}[X] = \frac{\theta}{2}.$$

2. The sample mean  $\bar{X}$  is given by:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

3. Equating the theoretical mean  $\mathbb{E}[X]$  to the sample mean  $\bar{X}$ :

$$\frac{\theta}{2} = \bar{X}.$$

Solving for  $\theta$ :

$$\hat{\theta}_2 = 2\bar{X}.$$

Hence,  $\hat{\theta}_2 = 2\bar{X}$  is the method of moments estimator for  $\theta$ .

**Part (b): Show that  $\hat{\theta}_1$  is a biased estimator and  $\hat{\theta}_2$  is an unbiased estimator of  $\theta$**

**Step 1: Bias of  $\hat{\theta}_1 = Y_n$**

- For  $Y_n = \max\{X_1, X_2, \dots, X_n\}$ , the expected value is:

$$\mathbb{E}[Y_n] = \frac{n}{n+1}\theta.$$

- Since  $\mathbb{E}[\hat{\theta}_1] = \mathbb{E}[Y_n] \neq \theta$ ,  $\hat{\theta}_1$  is a biased estimator.

**Step 2: Bias of  $\hat{\theta}_2 = 2\bar{X}$**

- The sample mean  $\bar{X}$  has an expectation:

$$\mathbb{E}[\bar{X}] = \frac{\theta}{2}.$$

- Substituting into  $\hat{\theta}_2 = 2\bar{X}$ :

$$\mathbb{E}[\hat{\theta}_2] = 2\mathbb{E}[\bar{X}] = 2 \cdot \frac{\theta}{2} = \theta.$$

- Since  $\mathbb{E}[\hat{\theta}_2] = \theta$ ,  $\hat{\theta}_2$  is an unbiased estimator.

**Part (c): Show that  $\hat{\theta}_3 = \frac{n+1}{n}\hat{\theta}_1$  is an unbiased estimator**

1. Substitute  $\hat{\theta}_1 = Y_n$  into  $\hat{\theta}_3$ :

$$\hat{\theta}_3 = \frac{n+1}{n}Y_n.$$

2. The expectation of  $\hat{\theta}_3$  is:

$$\mathbb{E}[\hat{\theta}_3] = \frac{n+1}{n}\mathbb{E}[Y_n].$$

3. From part (b),  $\mathbb{E}[Y_n] = \frac{n}{n+1}\theta$ . Substituting this:

$$\mathbb{E}[\hat{\theta}_3] = \frac{n+1}{n} \cdot \frac{n}{n+1}\theta = \theta.$$

4. Since  $\mathbb{E}[\hat{\theta}_3] = \theta$ ,  $\hat{\theta}_3$  is an unbiased estimator of  $\theta$ .

## Final Results

- (a)  $\hat{\theta}_2 = 2\bar{X}$  is a method of moments estimator.
- (b)  $\hat{\theta}_1 = Y_n$  is a biased estimator, while  $\hat{\theta}_2 = 2\bar{X}$  is an unbiased estimator.
- (c)  $\hat{\theta}_3 = \frac{n+1}{n}\hat{\theta}_1$  is an unbiased estimator of  $\theta$ .

## 5.3.10

Let  $X_1, \dots, X_{n_1}$  be a random sample from an  $N(\mu_1, \sigma^2)$  distribution and let  $Y_1, \dots, Y_{n_2}$  be a random sample from an  $N(\mu_2, \sigma^2)$  distribution.

Show that the pooled estimator:

$$\hat{\sigma}^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

is unbiased for  $\sigma^2$ , where  $S_1^2$  and  $S_2^2$  are the respective sample variances.

## Solution:

To show that the pooled estimator

$$\hat{\sigma}^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

is an unbiased estimator for  $\sigma^2$ , we proceed as follows:

## Step 1: Definitions

### 1. Sample Variances:

- For the sample  $X_1, \dots, X_{n_1}$ , the sample variance  $S_1^2$  is defined as:

$$S_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (X_i - \bar{X})^2.$$

The expected value of  $S_1^2$  is:

$$\mathbb{E}[S_1^2] = \sigma^2.$$

- Similarly, for the sample  $Y_1, \dots, Y_{n_2}$ , the sample variance  $S_2^2$  is defined as:

$$S_2^2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (Y_j - \bar{Y})^2.$$

The expected value of  $S_2^2$  is:

$$\mathbb{E}[S_2^2] = \sigma^2.$$

2. **Pooled Variance:** The pooled variance  $\hat{\sigma}^2$  combines the variances of the two samples:

$$\hat{\sigma}^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

## Step 2: Expected Value of $\hat{\sigma}^2$

To show that  $\hat{\sigma}^2$  is unbiased, we compute its expected value:

$$\mathbb{E}[\hat{\sigma}^2] = \mathbb{E} \left[ \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2} \right].$$

Using the linearity of expectation:

$$\mathbb{E}[\hat{\sigma}^2] = \frac{(n_1 - 1)\mathbb{E}[S_1^2] + (n_2 - 1)\mathbb{E}[S_2^2]}{n_1 + n_2 - 2}.$$

Substitute  $\mathbb{E}[S_1^2] = \sigma^2$  and  $\mathbb{E}[S_2^2] = \sigma^2$ :

$$\mathbb{E}[\hat{\sigma}^2] = \frac{(n_1 - 1)\sigma^2 + (n_2 - 1)\sigma^2}{n_1 + n_2 - 2}.$$

Factor out  $\sigma^2$ :

$$\mathbb{E}[\hat{\sigma}^2] = \sigma^2 \cdot \frac{(n_1 - 1) + (n_2 - 1)}{n_1 + n_2 - 2}.$$

Simplify the numerator:

$$(n_1 - 1) + (n_2 - 1) = n_1 + n_2 - 2.$$

Substitute back:

$$\mathbb{E}[\hat{\sigma}^2] = \sigma^2 \cdot \frac{n_1 + n_2 - 2}{n_1 + n_2 - 2}.$$

$$\mathbb{E}[\hat{\sigma}^2] = \sigma^2.$$

### Step 3: Conclusion

The expected value of  $\hat{\sigma}^2$  is equal to  $\sigma^2$ . Thus,  $\hat{\sigma}^2$  is an unbiased estimator of  $\sigma^2$ .

### Final Answer

The pooled estimator:

$$\hat{\sigma}^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

is unbiased for  $\sigma^2$ .

## 5.4.9

Let  $X_1, \dots, X_n$  be a random sample from an  $N(\mu, \sigma^2)$  distribution.

- Construct a  $(1 - \alpha)100\%$  confidence interval (CI) for  $\mu$  when the value of  $\sigma^2$  is known.
- Construct a  $(1 - \alpha)100\%$  confidence interval (CI) for  $\mu$  when the value of  $\sigma^2$  is unknown.

### Solution:

Let  $X_1, \dots, X_n$  be a random sample from an  $N(\mu, \sigma^2)$  distribution. The goal is to construct a  $(1 - \alpha)100\%$  confidence interval (CI) for  $\mu$  in two cases.

#### Case (a): When $\sigma^2$ is known

##### Step 1: Confidence Interval Formula

If the variance  $\sigma^2$  is known, the sampling distribution of the sample mean  $\bar{X}$  is:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$$

The  $(1 - \alpha) \times 100\%$  confidence interval for  $\mu$  is constructed using the standard normal distribution  $Z \sim N(0, 1)$ . The formula is:

$$\bar{X} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}},$$

where:

- $\bar{X}$  is the sample mean,
- $z_{\alpha/2}$  is the critical value from the standard normal distribution such that the area to the right of  $z_{\alpha/2}$  is  $\alpha/2$ ,
- $\sigma$  is the known population standard deviation,
- $n$  is the sample size.

### Step 2: Confidence Interval

The  $(1 - \alpha) \times 100\%$  confidence interval for  $\mu$  is:

$$\left( \bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right).$$

### Case (b): When $\sigma^2$ is unknown

#### Step 1: Use the $t$ -Distribution

When  $\sigma^2$  is unknown, we estimate it using the sample variance  $S^2$ . The sampling distribution of  $\bar{X}$  is no longer standard normal but follows a  $t$ -distribution with  $n - 1$  degrees of freedom:

$$\bar{X} \sim t_{n-1}.$$

The  $(1 - \alpha) \times 100\%$  confidence interval for  $\mu$  is:

$$\bar{X} \pm t_{\alpha/2, n-1} \cdot \frac{S}{\sqrt{n}},$$

where:

- $t_{\alpha/2, n-1}$  is the critical value from the  $t$ -distribution with  $n - 1$  degrees of freedom such that the area to the right of  $t_{\alpha/2, n-1}$  is  $\alpha/2$ ,
- $S$  is the sample standard deviation:

$$S = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}.$$

### Step 2: Confidence Interval

The  $(1 - \alpha) \times 100\%$  confidence interval for  $\mu$  is:

$$\left( \bar{X} - t_{\alpha/2, n-1} \cdot \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2, n-1} \cdot \frac{S}{\sqrt{n}} \right).$$

## Key Differences

1. When  $\sigma^2$  is known, the standard normal distribution ( $z$ -values) is used.
2. When  $\sigma^2$  is unknown, the  $t$ -distribution ( $t$ -values) with  $n - 1$  degrees of freedom is used.

### 5.5.4

Let the observed mean of a sample of size 45 be  $\bar{x} = 68.51$  from a distribution having variance 110. Find a 95% confidence interval (CI) for the true mean  $\mu$  and interpret the result and state any assumptions you have made.

## Solution:

We are given the following:

- Sample size:  $n = 45$ ,
- Sample mean:  $\bar{x} = 68.51$ ,
- Population variance:  $\sigma^2 = 110$ , so  $\sigma = \sqrt{110} \approx 10.488$ ,
- Confidence level: 95%.

## Step 1: Confidence Interval Formula

When the population variance ( $\sigma^2$ ) is known, the confidence interval for the population mean  $\mu$  is given by:

$$\text{CI} = \bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}},$$

where:

- $z_{\alpha/2}$  is the critical value from the standard normal distribution for a confidence level of 95% (i.e.,  $\alpha = 0.05$ ),
- $\sigma$  is the population standard deviation,
- $n$  is the sample size.

For 95% confidence,  $\alpha = 0.05$  and  $z_{\alpha/2} \approx 1.96$ .



## Step 2: Calculate the Margin of Error

The margin of error (ME) is calculated as:

$$\text{ME} = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}.$$

Substituting the given values:

$$\text{ME} = 1.96 \cdot \frac{10.488}{\sqrt{45}}.$$

First, compute  $\sqrt{45} \approx 6.708$ . Then:

$$\text{ME} = 1.96 \cdot \frac{10.488}{6.708} \approx 1.96 \cdot 1.563 \approx 3.064.$$

## Step 3: Confidence Interval

The confidence interval is:

$$\text{CI} = \bar{x} \pm \text{ME}.$$

Substituting  $\bar{x} = 68.51$  and  $\text{ME} = 3.064$ :

$$\text{CI} = 68.51 \pm 3.064.$$

This gives:

$$\text{Lower Bound} = 68.51 - 3.064 = 65.446,$$

$$\text{Upper Bound} = 68.51 + 3.064 = 71.574.$$

Thus, the 95% confidence interval is:

$$\text{CI} = (65.45, 71.57).$$

## Step 4: Interpretation

We are 95% confident that the true population mean  $\mu$  lies within the interval (65.45, 71.57).

## Step 5: Assumptions

The following assumptions are made:

- The sample is random and representative of the population.
- The population is normally distributed, or the sample size ( $n = 45$ ) is large enough for the Central Limit Theorem to apply.
- The population variance ( $\sigma^2$ ) is known.

### 5.5.13

Let  $\bar{X}$  be the mean of a random sample of size  $n$  from an  $N(\mu, 16)$  distribution. Find  $n$  such that:

$$P(\bar{X} - 2 < \mu < \bar{X} + 2) = 0.95.$$

#### **Solution:**

We are tasked with finding the sample size  $n$  such that the confidence interval

$$P(\bar{X} - 2 < \mu < \bar{X} + 2) = 0.95,$$

where  $\bar{X}$  is the sample mean from an  $N(\mu, 16)$  distribution.

#### **Step 1: Sampling Distribution of $\bar{X}$**

The sampling distribution of  $\bar{X}$  is:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right),$$

where  $\sigma^2 = 16$  is the population variance. The standard deviation of  $\bar{X}$  is:

$$\text{Standard Deviation of } \bar{X} = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{n}}.$$

#### **Step 2: Confidence Interval Relation**

For a 95% confidence level, the critical  $z$ -value is:

$$z_{\alpha/2} = 1.96.$$

The margin of error for the confidence interval is given by:

$$\text{Margin of Error} = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}.$$

Substituting the margin of error as 2:

$$2 = 1.96 \cdot \frac{4}{\sqrt{n}}.$$

#### **Step 3: Solve for $n$**

Rearranging the equation to solve for  $n$ :

$$\sqrt{n} = \frac{1.96 \cdot 4}{2},$$

$$\sqrt{n} = 3.92.$$

Squaring both sides:

$$n = 3.92^2 = 15.3664.$$

Since  $n$  must be an integer, round up to the next whole number:

$$n = 16.$$

### Final Answer

The required sample size is:

$$\boxed{n = 16}.$$

### Interpretation

With a sample size of  $n = 16$ , the sampling distribution of  $\bar{X}$  will have a small enough standard deviation such that the confidence interval  $(\bar{X} - 2, \bar{X} + 2)$  has a 95% probability of containing the true population mean  $\mu$ .

## 5.5.28

In a large university, the following are the ages of 20 randomly chosen employees:

24, 31, 28, 43, 28, 56, 48, 39, 52, 32,

38, 49, 51, 49, 62, 33, 41, 58, 63, 56.

Assuming that the data came from a normal population, construct a 95% confidence interval (CI) for the population mean  $\mu$  of the ages of the employees of this university. Interpret your answer.

### Solution:

We are tasked with constructing a 95% confidence interval (CI) for the population mean  $\mu$  based on the following data:

24, 31, 28, 43, 28, 56, 48, 39, 52, 32, 38, 49, 51, 49, 62, 33, 41, 58, 63, 56.

#### Step 1: Information given

- Sample size:  $n = 20$ ,
- Confidence level: 95% ( $\alpha = 0.05$ ).

## Step 2: Sample Mean and Sample Standard Deviation

**Sample Mean ( $\bar{x}$ ):** The sample mean is calculated as:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}.$$

Summing the data:

$$\sum_{i=1}^n x_i = 24 + 31 + 28 + \cdots + 56 = 884.$$

Thus:

$$\bar{x} = \frac{884}{20} = 44.2.$$

**Sample Standard Deviation ( $s$ ):** The sample standard deviation is calculated as:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}.$$

After computation (subtracting  $\bar{x} = 44.2$ , squaring deviations, summing, and dividing by  $n - 1 = 19$ ):

$$s \approx 12.15.$$

## Step 3: Confidence Interval Formula

When the population standard deviation is unknown, the confidence interval is given by:

$$\text{CI} = \bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}.$$

Where:

- $t_{\alpha/2, n-1} = t_{0.025, 19} \approx 2.093$  (from the  $t$ -table),
- $s \approx 12.15$ ,
- $n = 20$ .

Substitute these values:

$$\text{Margin of Error (ME)} = t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}} = 2.093 \cdot \frac{12.15}{\sqrt{20}}.$$

First, calculate  $\sqrt{20} \approx 4.472$ , then:

$$\text{ME} = 2.093 \cdot \frac{12.15}{4.472} \approx 2.093 \cdot 2.716 \approx 5.69.$$

### Step 4: Compute the Confidence Interval

The confidence interval is:

$$\text{CI} = \bar{x} \pm \text{ME}.$$

Substituting  $\bar{x} = 44.2$  and  $\text{ME} = 5.69$ :

$$\text{Lower Bound} = 44.2 - 5.69 = 38.51,$$

$$\text{Upper Bound} = 44.2 + 5.69 = 49.89.$$

Thus, the 95% confidence interval is:

$$\text{CI} = (38.51, 49.89).$$

### Step 5: Interpretation

We are 95% confident that the true mean age  $\mu$  of the employees in the university lies between 38.51 and 49.89.

### Assumptions

- The data is randomly selected and representative of the population.
- The data follows a normal distribution (or the Central Limit Theorem applies due to a reasonable sample size).

## 5.6.4

A random sample of 25 observations gave the following summary statistics:

$$\sum x_i = 234 \quad \text{and} \quad \sum x_i^2 = 3048.$$

Assuming that the data can be looked upon as a random sample from a normal population, construct a 95% confidence interval (CI) for the actual variance,  $\sigma^2$ .

### Solution:

We are tasked with constructing a 95% confidence interval (CI) for the population variance  $\sigma^2$  based on the given data:

$$\sum x_i = 234 \quad \text{and} \quad \sum x_i^2 = 3048, \quad \text{with sample size } n = 25.$$

### Step 1: Compute the sample variance $s^2$

The formula for the sample variance is:

$$s^2 = \frac{1}{n-1} \left( \sum x_i^2 - \frac{(\sum x_i)^2}{n} \right).$$

Substituting the given values:

$$s^2 = \frac{1}{25-1} \left( 3048 - \frac{(234)^2}{25} \right).$$

First, calculate  $\frac{(234)^2}{25}$ :

$$\frac{234^2}{25} = \frac{54756}{25} = 2190.24.$$

Now calculate:

$$3048 - 2190.24 = 857.76.$$

Thus:

$$s^2 = \frac{857.76}{24} \approx 35.74.$$

### Step 2: Confidence interval formula

The confidence interval for the population variance  $\sigma^2$  is given by:

$$\left( \frac{(n-1)s^2}{\chi_{1-\alpha/2}^2}, \frac{(n-1)s^2}{\chi_{\alpha/2}^2} \right),$$

where:

- $n-1 = 24$ ,
- $\alpha = 0.05$  for a 95% CI,
- $\chi_{1-\alpha/2}^2 = \chi_{0.975,24}^2 \approx 39.36$  (from chi-squared table),
- $\chi_{\alpha/2}^2 = \chi_{0.025,24}^2 \approx 12.40$ .

### Step 3: Calculate the bounds

The lower and upper bounds of the confidence interval are calculated as:

$$\text{Lower Bound} = \frac{(n-1)s^2}{\chi_{1-\alpha/2}^2} = \frac{(24)(35.74)}{39.36} = \frac{857.76}{39.36} \approx 21.8.$$

$$\text{Upper Bound} = \frac{(n-1)s^2}{\chi_{\alpha/2}^2} = \frac{(24)(35.74)}{12.40} = \frac{857.76}{12.40} \approx 69.2.$$

#### Step 4: Final Confidence Interval

The 95% confidence interval for the variance  $\sigma^2$  is:

$$\text{CI} = (21.8, 69.2).$$

#### Step 5: Interpretation

We are 95% confident that the true population variance  $\sigma^2$  lies between 21.8 and 69.2.

### Project

#### 5.10.9 Confidence intervals based on sampling distributions

If we want to obtain a  $(1 - \alpha)100\%$  CI for  $\theta$ , begin with an estimator  $\hat{\theta}$  of  $\theta$  and determine its sampling distribution. Now select two probability levels,  $\alpha_1$  and  $\alpha_2$ , so that  $\alpha = \alpha_1 + \alpha_2$ . Generally, we let  $\alpha_1 = \alpha_2$ . Take a sample and calculate the value of  $\hat{\theta}$ , say  $\hat{\theta} = k$ . Now we need to determine the values of the upper and lower confidence limits. Find a value  $\theta_L$  such that:

$$P(\hat{\theta} \geq k) = \alpha_1,$$

and  $\theta_U$  such that:

$$P(\hat{\theta} \leq k) = \alpha_2.$$

Then a  $(1 - \alpha)100\%$  CI for  $\theta$  will be:

$$\theta_L < \theta < \theta_U.$$

#### (a)

Let  $X_1, \dots, X_n$  be a random sample from a  $U(0, \theta)$  distribution. Obtain a  $(1 - \alpha)100\%$  CI for  $\theta$  using the method of sampling distribution.

#### (b)

Let  $X$  have a binomial distribution with parameters  $n$  and  $p$ . First, show that there is no quantity that satisfies the conditions of a pivotal quantity. Then, using the method of sampling distributions, obtain a  $(1 - \alpha)100\%$  CI for  $p$ .

## Solution:

## Confidence Intervals Based on Sampling Distributions

### Problem Statement

(a) Let  $X_1, X_2, \dots, X_n$  be a random sample from a  $U(0, \theta)$  distribution. Obtain a  $(1 - \alpha)100\%$  confidence interval (CI) for  $\theta$  using the method of sampling distribution.

(b) Let  $X$  have a binomial distribution with parameters  $n$  and  $p$ . First, show that there is no quantity that satisfies the conditions of a pivotal quantity. Then, using the method of sampling distributions, obtain a  $(1 - \alpha)100\%$  CI for  $p$ .

### Solution

#### Part (a): Confidence Interval for $\theta$ in $U(0, \theta)$

**Step 1: Distribution of  $Y_n$**  Let  $Y_n = \max(X_1, X_2, \dots, X_n)$ . The cumulative distribution function (CDF) of  $Y_n$  is given by:

$$F_{Y_n}(y) = P(Y_n \leq y) = P(X_1 \leq y, X_2 \leq y, \dots, X_n \leq y).$$

Since  $X_i \sim U(0, \theta)$  are independent and identically distributed (i.i.d.):

$$F_{Y_n}(y) = (P(X_1 \leq y))^n = \left(\frac{y}{\theta}\right)^n, \quad \text{for } 0 \leq y \leq \theta.$$

The probability density function (PDF) is obtained by differentiating  $F_{Y_n}(y)$ :

$$f_{Y_n}(y) = \frac{d}{dy} F_{Y_n}(y) = \frac{ny^{n-1}}{\theta^n}, \quad 0 \leq y \leq \theta.$$

**Step 2: Confidence Interval for  $\theta$**  Using the sampling distribution of  $Y_n$ , we derive a  $(1 - \alpha)100\%$  confidence interval. Define:

$$P\left(\frac{Y_n}{\theta} \leq k\right) = 1 - \alpha, \quad \text{where } k = (1 - \alpha)^{1/n}.$$

Rewriting this in terms of  $\theta$ :

$$\theta \geq \frac{Y_n}{k}, \quad \text{or equivalently, } \theta \geq \frac{Y_n}{(1 - \alpha)^{1/n}}.$$

Thus, the confidence interval is:

$$\theta \in \left[ \frac{Y_n}{(1 - \alpha)^{1/n}}, \infty \right).$$



**Part (b): Confidence Interval for  $p$  in Binomial Distribution**

**Step 1: Sampling Distribution of  $\hat{p}$**  Let  $X \sim \text{Binomial}(n, p)$ , where the sample proportion is  $\hat{p} = \frac{X}{n}$ . For large  $n$ , the sampling distribution of  $\hat{p}$  is approximately normal:

$$\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right).$$

**Step 2: Confidence Interval** Using the pivotal quantity:

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}},$$

where  $Z \sim N(0, 1)$ , the  $(1 - \alpha)100\%$  confidence interval is derived as:

$$P\left(-Z_{\alpha/2} \leq \frac{\hat{p} - p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \leq Z_{\alpha/2}\right) = 1 - \alpha.$$

Simplifying:

$$\hat{p} - Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \leq p \leq \hat{p} + Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

**Final Confidence Interval for  $p$ :**

$$p \in \left[ \hat{p} - Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}, \hat{p} + Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right].$$